Credit Risk Portfolio Analytics – Take-Home Assignment (v2)

Oct 27, 2026 · @Essex User 01

Overview

Build a working credit-risk monitoring dashboard with two embedded AI features — an LLM "Ask the Portfolio" assistant and anomaly detection — from 5 synthetic bank tables, and submit it in the standard format in section 5. You have **3 calendar days** from receiving this brief.

Data provided (synthetic, ~10,000 customers; see `DATA_DICTIONARY.md` in the data pack):

Table	Grain	Approx. rows	Key
`customers.csv`	1 row per customer	10,000	`customer_id`
`loan_accounts.csv`	1 row per loan; holds `target_default`	14,700	`account_id` → `customer_id`
`bureau_credit.csv`	1 row per external credit line	23,000	`customer_id`
`repayment_history.csv`	1 row per account per month	176,000	`account_id`
`collections_events.csv`	1 row per collections action	8,600	`account_id`

You may use any tools, libraries and AI coding assistants. You must be able to explain every part of your solution in the follow-up interview.

Required build

1. **Data ingestion:** load and join all 5 tables, handle the data-quality issues, and produce one clean, analysis-ready dataset. Document your join logic and every data-quality decision.

2. **Portfolio analytics:** profile the portfolio across income, employment type, region, age and loan type. Give 4–5 insights, each with a "so what", including at least one cross-dimensional cut.
3. **Default model:** predict `target_default` using engineered features from the bureau and repayment history, with a logistic regression baseline and one ML model. Report AUC, KS, Gini and feature importance or SHAP, and explain how you would monitor the model after deployment.
4. **Dashboard:** an interactive dashboard with a portfolio view (risk segments, exposure, trends across dimensions), an account-level view (risk score and key drivers), at least one working filter or drill-down, and both AI features (section 3) embedded in it.

Mandatory AI features

Build both features, embedded in the dashboard and working on the hosted link.

1. **"Ask the Portfolio" (LLM):** natural-language Q&A over the dashboard data using an LLM, grounded in the actual underlying data — no hallucinated numbers.
   - *Expected:* an analyst types a question in the dashboard (e.g. "Which loan type has the highest default rate?") and gets a short plain-English answer with the exact figures and where they came from. Questions the data cannot answer are declined.
2. **Anomaly detection:** flag accounts or customers with unusual repayment or bureau patterns compared with their own history or their peer segment.
   - *Expected:* a review list of flagged accounts in the dashboard. For each account, the analyst can see why it was flagged and record a decision (e.g. confirm or dismiss).

For each feature, state in the dashboard what triggers it, what a false positive and a false negative look like, and the human-in-the-loop step.

Documents

Submit exactly two documents, each **one page, as PDF**. No slide deck is needed.

1. **Architecture diagram** (`docs/architecture.pdf`)
   - The flow from raw data → ingestion → analytics → model → dashboard, with both AI features shown.
   - The tool or library used at each stage, where the LLM is called, and what data is sent to it.
2. **Solution overview** (`docs/solution_overview.pdf`)

- What the dashboard shows and how a risk analyst would use it, view by view.
- The two AI features: how each works, its trigger, false positive, false negative and human-review step.
- Model results (AUC, KS, Gini) and your modelling approach.
- Your top 3 portfolio insights.
- Key assumptions, and what you would need from a bank's IT and data teams to put this into production.

Content beyond one page will not be read.

Submission standard

Submit one email containing three links. Do not attach zip files.

Item	Requirement
1. Hosted dashboard link	A live app on Streamlit Community Cloud, Hugging Face Spaces or similar (both are free and have a secrets store for your API key). It must open without a login, with the LLM feature working, and stay live for **14 days** after submission. A static HTML page cannot hold an API key safely, so it does not qualify on its own.
2. GitHub repository link	Public, or private with access given to the reviewer account we name. Must match the hosted app and contain no API keys or other secrets.
3. Demo video link	3–5 minutes (Loom, YouTube unlisted or a shared drive). Show the portfolio and account views and both AI features in use.

Repository layout

```
README.md
requirements.txt        # every package pinned, e.g. pandas==2.2.3
.env.example            # names of required keys, no values
app/app.py              # dashboard entry point
src/                    # ingestion, analytics, model, AI features
data/raw/               # the 5 CSVs as provided
docs/architecture.pdf
docs/solution_overview.pdf
tests/                  # optional; recommended
```

README must include

- The Python version used.
- Run instructions as no more than three commands: install, build the data and model, start the dashboard.
- How to set the LLM API key, and what the app does if no key is set.
- Where each item in this brief lives in the repo.